

MedForge Cheat Sheet — Acromegaly

Educational draft - source review required

Acromegaly – Cheat Sheet [S1][S2][S4]

Definition: Acromegaly is a rare endocrine disorder caused by hypersecretion of growth hormone (GH), most commonly due to a pituitary adenoma [S1][S4].

Mechanism Chain:

GH → stimulates insulin-like growth factor 1 (IGF-1) → increased peripheral tissue growth and metabolic disruption → insulin resistance → impaired glucose uptake → diabetes mellitus [S1][S3][S4].

Must-Know Facts:

- Diagnosis confirmed by elevated serum IGF-1 and lack of GH suppression after glucose tolerance test [S4].
- Pituitary MRI is advised to identify adenoma [S4].
- Diabetes mellitus occurs in ~1/3 of patients, independent of obesity, with lean phenotype and reduced adipose tissue [S3].
- Cardiovascular morbidity and mortality are increased in acromegaly [S3].
- Elderly patients show rising incidence due to increased life expectancy; diagnosis is delayed due to symptom overlap with aging [S5].
- Long-term GH exposure may increase risk of benign or malignant tumors; association with cancer remains debated [S6].

Red-Flag Confusions:

- Acromegaly is not caused by insulin-like growth factor 1 overproduction; GH drives IGF-1 elevation [S1][S4].
- Insulin resistance in acromegaly occurs without obesity, unlike typical type 2 diabetes [S3].
- Growth hormone levels at last follow-up and IGF-1 at diagnosis are independent predictors of mortality [S6].

Quick Recall Prompts:

- “GH → IGF-1 → insulin resistance → diabetes” [S3][S4].
- “MRI first, then surgery” [S4].
- “Diabetes in acromegaly: lean, not obese” [S3].
- “Older patients: more comorbidities, worse prognosis” [S5].
- “Cancer link? Debated; monitor closely” [S6].

Note: Management of diabetes in acromegaly lacks specific guidelines due to limited research [S3]. [S3]

References

[S1] PubMed PMID 11571931 — Acromegaly. <https://pubmed.ncbi.nlm.nih.gov/11571931/>

[S2] PubMed PMID 9429856 — Acromegaly. <https://pubmed.ncbi.nlm.nih.gov/9429856/>

[S3] PubMed PMID 38844688 — Diabetes mellitus in patients with acromegaly: pathophysiology, clinical challenges and management. <https://pubmed.ncbi.nlm.nih.gov/38844688/>

[S4] PubMed PMID 35120696 — Diagnosis and Treatment of Acromegaly: An Update.
<https://pubmed.ncbi.nlm.nih.gov/35120696/>

[S5] PubMed PMID 31939489 — Acromegaly in the elderly patient.
<https://pubmed.ncbi.nlm.nih.gov/31939489/>

[S6] PubMed PMID 38007382 — Acromegaly and Cancer: An Update.
<https://pubmed.ncbi.nlm.nih.gov/38007382/>